# Robust Foreground Estimation via Structured Gaussian Scale Mixture Modeling

Guangming Shi, *Senior Member, IEEE*, Tao Huang, Weisheng Dong, *Member, IEEE*, Jinjian Wu, and Xuemei Xie

***Abstract*— Recovering the background and foreground parts from video frames has important applications in video surveillance. Under the assumption that the background parts are stationary and the foreground are sparse, most of existing methods are based on the framework of robust principal component analysis (RPCA), i.e., modeling the background and foreground parts as a low-rank and sparse matrices, respectively. However, in realistic complex scenarios, the conventional $\ell_1$ norm sparse regularizer often fails to well characterize the varying sparsity of the foreground components. How to select the sparsity regularizer parameters adaptively according to the local statistics is critical to the success of the RPCA framework for background subtraction task. In this paper, we propose to model the sparse component with a Gaussian scale mixture (GSM) model. Compared with the conventional $\ell_1$ norm, the GSM-based sparse model has the advantages of jointly estimating the variances of the sparse coefficients (and hence the regularization parameters) and the unknown sparse coefficients, leading to significant estimation accuracy improvements. Moreover, considering that the foreground parts are highly structured, a structured extension of the GSM model is further developed. Specifically, the input frame is divided into many homogeneous regions using superpixel segmentation. By characterizing the set of sparse coefficients in each homogeneous region with the same GSM prior, the local dependencies among the sparse coefficients can be effectively exploited, leading to further improvements for background subtraction. Experimental results on several challenging scenarios show that the proposed method performs much better than most of existing background subtraction methods in terms of both performance and speed.**

***Index Terms*— Background subtraction, Gaussian scale mixture, stochastic optimization, alternative minimization.**

## I. INTRODUCTION

DETECTING moving objects from video frames has many important applications, such as surveillance tracking [1], [2], scene understanding [3], [4], vehicle navigation [5], augmented reality [6] and Magnetic particle imaging [7]. As one of the most widely used approaches for moving objects detection, background subtraction aims to separate the moving objects from the background images and generates the mask of the moving objects. Due to the issues of dynamic backgrounds, irregular objects motion and bad weather, it is rather challenging to robustly separate the foregrounds from the backgrounds.

In the past decades, various background subtraction methods have been proposed. Pixel-based methods [8]–[13] try to learn background or foreground models to classify the pixels using only the pixel intensities. Classic pixel-based method are those using Gaussian Mixture Model (GMM) [9], [14], where each pixel is represented as a mixture of weighted Gaussian models. Pixels that cannot be well described by the GMM model are classified as foreground pixels. Improved GMM-based methods try to automatically learn the number of Gaussian components [15]–[17] such that the GMM model can better adapt to the scene. To exploit local dependencies between neighboring pixels, local binary pattern (LBP) model has been proposed in [18] for better background representation and led to better performance. Furthermore, neural networks based methods have also been developed [10], [19], which learn the background model adaptively and can better deal with the complex scenarios containing moving backgrounds and illumination changes. Despite the efficiency of these pixel-based methods, they are usually built based on some restrict assumptions and learn from training datasets. Moreover, those methods cannot well describe the global correlations of the background images across different frames. For complex scenarios, the performances of the pixel-based models are likely to degrade or even fail due to the changes of illuminations and perspectives and the dynamic background.

Considering that the background images across different frames are highly correlated, the Robust Principal Component Analysis (RPCA) technique [20] has been used for background subtraction. In the RPCA-based methods [21]–[27], each frame is first converted to be a column of the data matrix and then the resulting matrix is decomposed into a low-rank background matrix and a sparse foreground matrix. As the RPCA-based methods can fully exploit the temporal correlations of the backgrounds, these methods have attracted a lot of attentions. Despite the powerful low-rank model for describing the correlations of the backgrounds, the classic $\ell_1$ sparse model, corresponding to the independent and identical distributed (i.i.d) Laplace model, cannot accurately model the moving objects for different scenarios. Thus, the original RPCA method cannot achieve satisfactory performance for complex scenarios. To improve the performance of the RPCA model, other priors of the moving objects have been

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The authors are with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: wsdong@mail.xidian.edu.cn).

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incorporated. In [21], the Markov Random Field (MRF) model is used to model the contiguous sparse supports explicitly, and the $\ell_0$ norm is used to regularize the sparse foreground parts. However, the MRF model tends to generate over-smoothed supports of the foreground objects. Other improved RPCA methods tried to solve the scale issue problem, i.e., the universal $\ell_1$ regularization parameter cannot handle the foreground objects of different size. In [22] and [23], two-pass RPCA methods have been proposed. In the first pass, these methods first detect the possible regions of the foreground parts by conducting the first-pass RPCA and also estimate the motion saliency using optical flow. Based on the initially detected foregrounds and the motion saliency, these method tune the regularization parameters for block sparsity for better performance. Obviously, the computational complexity of the two-pass methods is very high. Recently, Cao *et al.*, propose to combine total variation (TV) regularization with the RPCA model to address the issues of the lingering objects and the dynamic backgrounds [24]. By exploiting the spatial and temporal continuity of foreground, the TV regularized RPCA model outperforms the conventional RPCA model. However, this method still cannot well solve the scale problem, as it still uses the global regularization parameter for the $\ell_1$ norm. For a more comprehensive review of the RPCA-based background subtraction methods, please refer to [28].

To exploit the structural correlations among neighboring pixels of video frames, spatially-consistent approaches have also been developed. In [29], superpixel-based online matrix decomposition method was proposed for more reliable low-rank decomposition and a generalized fussed LASSO was also adopted to exploit structural correlations of the foreground components. In [30], the minimum spanning tree (MST) based low-rank learning has been proposed. By imposing smoothness constrained on similar pixels that were found by the MST, the spatially-consistent low-rank model can be improved. To improve the robustness of the low-rank model against outliers, in [31], a spatial and temporal graph regularization was incorporated into the low-rank matrix completion framework for improved segmentation of background. Considering that real-world backgrounds usually span multiple manifolds, spatialtemporal sparse spectral clustering regularizer for RPCA has also been proposed in [32] for efficient background modeling. By enforcing continuity on the low-dimensional multiple manifolds, this method can better handle dynamic backgrounds, occlusions, and intensity variations.

In addition to the matrix-based RPCA method, the tensor-based RPCA background subtraction methods [29], [33]–[35] also have been proposed, where the observed video frames are represented as a three-dimensional cubic. In these methods, the well-known Tucker decomposition technique is used to unfold the tensor along each mode to be a low-rank matrix. Compared with the matrix-based RPCA methods, the tensor-based RPCA methods have the advantages in fully exploiting the spatial-temporal correlations and thus can lead to superior performance. However, the rank of each unfolding matrix is still required as a prior, which is varying for different scenarios and tends to be overestimated or underestimated. Moreover, the computational complexity, as well as the memory consumption of the Tucker decomposition is very high or large, and grows rapidly with the size of the video. Note that to reduce the computational complexity of the low-rank approximation, some attempts have been made to develop fast singular value decomposition (SVD) technique [36]–[38], which can be adopted in the RPCA framework.

In this paper, to address the scale issue of the RPCA framework, we propose to model each pixel of the moving objects with a Gaussian Scale Mixture (GSM) model [39], [40]. The basic idea is to model each sparse foreground pixel as a product of a standard Gaussian random variable and a positive scaling variable and impose a hyperprior (i.e., the Jeffrey prior [41] used in this work) over the positive scaling variables. Compared with the classic Laplace model for $\ell_1$-norm regularization, where the parameters of the Laplace model have to be manually optimized, the GSM model allows us to jointly estimate both the sparse pixels and the scaling variables from the observed data under the MAP estimation framework via the method of alternative optimization. Moreover, to exploit the correlations among the neighboring pixels, we further extend the GSM models into structured GSM models by modeling the set of pixels belonging to an object with the same positive scaling variable. To this end, a super-pixel based segmentation method is first employed to segment the frames and group the pixels for each homogenous regions. By exploiting the correlations between the neighboring pixels, we can further improve the foreground estimation performance. In addition, to avoid the high computational complexity of the SVD in the RPCA framework, we adopt the stochastic optimization strategy for the low-rank approximation, which can significantly reduce the computational complexity. Experimental results on three challenging scenarios, including dynamic background, irregular moving objects and bad weather, show that the proposed method outperforms existing state-of-the-art background subtraction methods.

The rest of the paper is organized as follows. Section II introduces the background of the RPCA framework for background subtraction. The proposed structured GSM model based background subtraction framework is presented in Section III, followed by the proposed optimization algorithm for the proposed method. Section V presents the experimental results on several challenging datasets. Section VI concludes the paper.

## II. Related Works

In this section, we briefly review the works related to the proposed GSM-based background substraction method, i.e., the RPCA-based background subtraction methods and the scale mixture models that have been used in the applications of image restoration.

In the RPCA-based background subtraction methods, the observed $N$ video frames of size $m \times n$ formulated as matrix $\boldsymbol{D}$, is decomposed as $\boldsymbol{D} = \boldsymbol{L} + \boldsymbol{S}$, where $\boldsymbol{D} \in \mathbb{R}^{p\times N} = \{\boldsymbol{d}_1, \ldots, \boldsymbol{d}_N\}$ is formed by vectorizing each frame as a column of the matrix, $\boldsymbol{L} \in \mathbb{R}^{p\times N} = \{\boldsymbol{l}_1, \ldots, \boldsymbol{l}_N\}$ and $\boldsymbol{S} \in \mathbb{R}^{p\times N} = \{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_N\}$ denote the background and foreground components, respectively, and $p = m \times n$. Denote by $\boldsymbol{d}_i$ the $i$th column of the matrix $\boldsymbol{D}$ and by $\boldsymbol{D}_{i,t}$ the $i$th pixel at the $t$th frame $\boldsymbol{d}_t$. Since the background images have high correlations, the background matrix $\boldsymbol{L}$ should be low-rank, whereas the foreground matrix $\boldsymbol{S}$ containing moving objects, e.g., cars or pedestrians are usually sparse. Thus, the estimation of $\boldsymbol{L}$ and $\boldsymbol{S}$ from $\boldsymbol{D}$ can be casted as the Robust Principal Component Analysis (RPCA) problem [20], i.e.,

$$\min_{\boldsymbol{L},\boldsymbol{S}} \ \mathrm{rank}(\boldsymbol{L}) + \lambda\|\boldsymbol{S}\|_0, \quad s.t.\ \boldsymbol{D} = \boldsymbol{L} + \boldsymbol{S}, \tag{1}$$

where rank$(\cdot)$ denotes the rank of a matrix. As both the rank and the $\ell_0$ of Eq. (1) are nonconvex and NP hard, they are often replaced with their convex surrogates, i.e.,

$$\min_{\boldsymbol{L},\boldsymbol{S}} \ \|\boldsymbol{L}\|_* + \lambda\|\boldsymbol{S}\|_1, \quad s.t.\ \boldsymbol{D} = \boldsymbol{L} + \boldsymbol{S}, \tag{2}$$

where $\|\cdot\|_*$ is the nuclear norm (sum of the singular values). By selecting an appropriate Lagrange multipliers, the constrained optimization problem can be converted into a non-constrained one, i.e.,

$$\min_{\boldsymbol{L},\boldsymbol{S}} \ \|\boldsymbol{D} - \boldsymbol{L} - \boldsymbol{S}\|_2^2 + \eta\|\boldsymbol{L}\|_* + \lambda\|\boldsymbol{S}\|_1, \tag{3}$$

which can be solved by alternative optimization. In the RPCA model, the selection of the parameters is critical to the success of this model. However, due to the scale issue of the moving objects, it is impossible to choose a global regularization parameter $\lambda$ that are suitable for all moving objects. Moreover, the $\ell_1$ norm sparse regularization cannot exploit the correlations between the neighboring pixels. To overcome the scale problem, two-stage RPCA methods [22], [23] have been proposed, which first roughly detect the moving objects and then set the regularization parameters based on the estimated motion saliency of foregrounds. However, the performance of two-stages methods largely depend on the accuracy of the initial detection of the foregrounds and still cannot achieve satisfying performance. Also, the computational complexity of these methods are very high.

As a classic statistical model, the Gaussian scale mixture (GSM) models [39], [40] have been used to model the wavelet coefficients or sparse coding coefficients for image restoration [42], [43]. In [42], the GSM model was used to model the local dependencies of wavelet coefficients, and a Bayesian least square estimator was developed for wavelet image denoising. Similar to the GSM model, in [44] the Laplacian scale mixture model has been proposed to model the dependencies among sparse coding coefficients for compressive sensing recovery. By imposing a Gamma distribution prior over the scale parameter, the scale parameters can be jointly updated along with the sparse codes. However, the update of the scale parameter is quite similar to that of the regularization parameter of the reweighted $\ell_1$-norm sparsity term [45]. Recently, Dong *et al.*, propose to model the sparse representation coefficients with GSM models for image restoration applications [43]. By modeling the scale parameters with the Jeffrey prior [41], both the scale parameters and the sparse coefficients were jointly estimated from the observed image by alternatively solving two sub-problems. Furthermore, by characterizing the sparse coefficients of a set of similar image patches with the same scale parameters, the nonlocal self-similarity prior can be effectively exploited. Similarly, in [46] the Laplacian scale mixture (LSM) model has also been proposed for mixed noise removal, where the impulse noise were characterized by the scale mixture models. Inspired by the success of the GSM/LSM modeling for sparse representation coefficients, in this paper we propose to model the sparse moving objects with the GSM models, which can well overcome the scale-issue of RPCA model.

## III. Foreground Estimation via Structured Gaussian Scale Mixture Modeling

### A. Foreground Estimation via Gaussian Scale Mixture Modeling

In this section, we first propose a Maximum a Posterior (MAP) estimator for estimating $\boldsymbol{L}$ and $\boldsymbol{S}$ from the observed $\boldsymbol{D}$. Using the MAP estimator, the estimation of $\boldsymbol{L}$ and $\boldsymbol{S}$ from $\boldsymbol{D}$ can be formulated as

$$\begin{aligned}(\boldsymbol{L}, \boldsymbol{S}) &= \mathrm{argmax} \log P(\boldsymbol{D}|\boldsymbol{L}, \boldsymbol{S})P(\boldsymbol{L}, \boldsymbol{S}) \\ &= \mathrm{argmax} \log P(\boldsymbol{D}|\boldsymbol{L}, \boldsymbol{S}) + \log P(\boldsymbol{L}, \boldsymbol{S}) \\ &= \mathrm{argmax} \log P(\boldsymbol{D}|\boldsymbol{L}, \boldsymbol{S}) + \log P(\boldsymbol{L}) + \log P(\boldsymbol{S}),\end{aligned} \tag{4}$$

where we have used the fact that the background component $\boldsymbol{L}$ and the foreground component $\boldsymbol{S}$ are nearly independent. The likelihood term $P(\boldsymbol{D}|\boldsymbol{L}, \boldsymbol{S})$ can be characterized by the zero-mean Gaussian distribution with variance $\sigma_w^2$, i.e.,

$$P(\boldsymbol{D}|\boldsymbol{L}, \boldsymbol{S}) = \frac{1}{\sqrt{2\pi}\sigma_w} exp(-\frac{\|\boldsymbol{D} - \boldsymbol{L} - \boldsymbol{S}\|_F^2}{2\sigma_w^2}). \tag{5}$$

Regarding the prior model of $\boldsymbol{L}$, it can be expressed as

$$P(\boldsymbol{L}) \propto \frac{1}{c}\exp(-\eta\|\boldsymbol{L}\|_*), \tag{6}$$

where $c$ is constant. It is easy to verify that if $P(\boldsymbol{S})$ is modeled with an independent and identically distributed (i.i.d) zero-mean Laplacian model, i.e., $P(S_{i,t}) \propto \frac{1}{2\theta_{i,t}}\exp(-\frac{|S_{i,t}|}{\theta_{i,t}})$, the MAP of Eq.(4) can be expressed as

$$(\boldsymbol{L}, \boldsymbol{S}) = \underset{\boldsymbol{L},\boldsymbol{S}}{\mathrm{argmin}} \ \frac{1}{2}\|\boldsymbol{D}-\boldsymbol{L}-\boldsymbol{S}\|_F^2 + \eta\sigma_w^2\|\boldsymbol{L}\|_* + \sum_{i,t}\lambda_{i,t}|S_{i,t}|, \tag{7}$$

where $\lambda_{i,t} = \sigma_w^2/\theta_{i,t}$, $\theta_{i,t}$ is the standard deviation of $S_{i,t}$. In realistic scenarios, $\theta_{i,t}$ of each $S_{i,t}$ are unknown and it is not easy to estimate them from the observations, as $\boldsymbol{S}$ is also unknown.

In this paper, we propose to model each foreground pixels with the Gaussian Scale Mixture (GSM) models. In GSM modeling, each pixel $S_{i,t}$ is decomposed into a product of a random Gaussian variable $\alpha_{i,t}$ and a positive hidden multiplier $\theta_{i,t}$, i.e., $S_{i,t} = \theta_{i,t}\alpha_{i,t}$. Each foreground pixel $S_{i,t}$ is then modeled as a zero-mean Gaussian distribution of standard deviation $\theta_{i,t}$. By imposing a prior distribution $P(\theta_{i,t})$ over $\theta_{i,t}$ and assuming that $\theta_{i,t}$ and $S_{i,t}$ are indenpendent, the GSM modeling of $\boldsymbol{S}$ can be formulated

$$P(\boldsymbol{S}) = \prod_{i,t} P(S_{i,t}), \quad P(S_{i,t}) = \int_0^\infty P(S_{i,t}|\theta_{i,t})P(\theta_{i,t})d\theta_{i,t}. \tag{8}$$

It has been known that the GSM model can well express several sparse distributions, such as Laplacian, Generalized Gaussian, and student's t-distributions, for an appropriate $P(\theta_{i,t})$ [39]. In our previous work [43], the GSM model has been successfully used to model the sparse coefficients for sparse coding. Here, we propose to use GSM to characterize the sparse moving objects.

Generally, for most choices of $P(\theta_{i,t})$, there is no analytical expression of $P(S_{i,t})$ and thus it is difficult to compute $P(S_{i,t})$ with MAP. However, such difficulty can be avoided by joint estimation of $P(S_{i,t}, \theta_{i,t})$. By replacing $P(S_{i,t})$ with $P(S_{i,t}, \theta_{i,t})$ in the MAP estimator of Eq. (4), we obtain

$$\begin{aligned}(\boldsymbol{L}, \boldsymbol{S}, \boldsymbol{\Theta}) &= \operatorname{argmax} \log P(\boldsymbol{D}|\boldsymbol{L}, \boldsymbol{S})P(\boldsymbol{L}, \boldsymbol{S}, \boldsymbol{\Theta}) \\ &= \operatorname{argmax} \log P(\boldsymbol{D}|\boldsymbol{L}, \boldsymbol{S}) + \log P(\boldsymbol{L}, \boldsymbol{S}, \boldsymbol{\Theta}) \\ &= \operatorname{argmax} \log P(\boldsymbol{D}|\boldsymbol{L}, \boldsymbol{S}) + \log P(\boldsymbol{L}) \\ &\quad + \log P(\boldsymbol{S}|\boldsymbol{\Theta}) + \log P(\boldsymbol{\Theta}), \end{aligned} \tag{9}$$

where $\boldsymbol{\Theta} = [\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \cdots, \boldsymbol{\theta}_N] \in \mathbb{R}^{p\times N}$ denotes the matrix of the positive multipliers. By substituting the prior of $P(\boldsymbol{L})$ and the GSM prior of $P(\boldsymbol{S}, \boldsymbol{\Theta})$ into this MAP estimator, we can obtain the following objective function

$$\begin{aligned}(\boldsymbol{L}, \boldsymbol{S}, \boldsymbol{\Theta}) = \underset{\boldsymbol{L},\boldsymbol{S},\boldsymbol{\Theta}}{\operatorname{argmin}} \ & \frac{1}{2\sigma_w^2}\|\boldsymbol{D} - \boldsymbol{L} - \boldsymbol{S}\|_F^2 + \eta\|\boldsymbol{L}\|_* \\ & + \sum_t \sum_i \frac{S_{i,t}^2}{2\theta_{i,t}^2} + 2\sum_t \sum_i \log \theta_{i,t}, \end{aligned} \tag{10}$$

where we assumed that each pixel $S_{i,t}$ are independent and the noninformative Jeffrey's prior $P(\theta_{i,t}) = \frac{1}{\theta_{i,t}}$ [41] is used. Note that $\boldsymbol{S} = \boldsymbol{\Theta} \odot \boldsymbol{A}$, where $\odot$ denotes the pixel-wise product and $\boldsymbol{A} = [\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \cdots, \boldsymbol{\alpha}_N] \in \mathbb{R}^{p\times N}$ is the matrix representation of the Gaussian random variables $\alpha_{i,t}$. Since the Jeffrey's prior $P(\theta_{i,t}) = \frac{1}{\theta_{i,t}}$ is numerical unstable, we introduce a small constant $\epsilon$ into $P(\theta_i)$, as $P(\theta_i) = \frac{1}{(\theta_i+\epsilon)}$. Then, Eq. (10) can be rewritten as

$$\begin{aligned}(\boldsymbol{L}, \boldsymbol{\Theta}, \boldsymbol{A}) = \underset{\boldsymbol{L},\boldsymbol{\Theta},\boldsymbol{A}}{\operatorname{argmin}} \ & \|\boldsymbol{D} - \boldsymbol{L} - \boldsymbol{\Theta} \odot \boldsymbol{A}\|_F^2 + 2\eta\sigma_w^2\|\boldsymbol{L}\|_* \\ & + 4\sigma_w^2 \sum_t \sum_i \log(\theta_{i,t} + \epsilon) + \sigma_w^2\|\boldsymbol{A}\|_F^2. \end{aligned} \tag{11}$$

### B. Foreground Estimation via Structured Gaussian Scale Mixture Modeling

In the proposed objective function of Eq. (11), each foreground pixel $S_{i,t}$ are assumed to be independent and identically distributed. However, it is well-known that the neighboring pixels often have strong correlations. Pixels belonging to the same object or homogeneous region should be characterized with the same prior, i.e., the same $\theta_{i,t}$. To exploit the correlations between the neighboring pixels, we further extend the i.i.d GSM model of Eq. (8) into structured GSM model, i.e.,

$$\begin{aligned} P(\boldsymbol{S}) &= \prod_t \prod_k \prod_{j\in\mathcal{G}_k} P(S_{j,t}), \\ P(S_{j,t}) &= \int_0^\infty P(S_{j,t}|\theta_{k,t})P(\theta_{k,t})d\theta_{k,t}, \quad j \in \mathcal{G}_{k,t}, \end{aligned} \tag{12}$$

Fig. 1. An example of ERS segmentation for an input frame.

where we have divided each foreground frame $\boldsymbol{s}_t$ into $K$ segments, denoted as $\mathcal{G}_{k,t}$. Each segment is recognized as a homogeneous region. Then, the foreground pixels in each $\mathcal{G}_{k,t}$ are characterized by the same second-order statistics $\theta_{k,t}$. A practical issue of Eq. (12) is that the foreground frame $\boldsymbol{s}_t$ is unknown, and thus we cannot obtain the segmentation and compute the prior of Eq. (12). To avoid such difficulty, we propose to obtain the segmentation using the observed frame $\boldsymbol{d}_t$. In this paper, we use the efficient Entropy Rate superpixel Segmentation (ERS) method [47] to segment each input frame $\boldsymbol{d}_t$ into $K$ homogeneous regions. An example of the segmentation is shown in Fig. 1, from which we can see that the pixels in the observed frame can be well segmented into different homogeneous regions. Based on the pixel grouping using $\boldsymbol{d}_t$, we can group the foreground pixels into $K$ groups, denoted as $\mathcal{G}_{k,t}$.

By substituting the structured GSM model of Eq. (12) into the MAP estimator of Eq. (9), the structured GSM model based foreground modeling can be formulated as

$$\begin{aligned}(\boldsymbol{L}, \boldsymbol{\Theta}, \boldsymbol{A}) = \underset{\boldsymbol{L},\boldsymbol{\Theta},\boldsymbol{A}}{\operatorname{argmin}} \ & \sum_t \sum_k \sum_{j\in\mathcal{G}_{k,t}} (D_{j,t} - L_{j,t} - \theta_{k,t}\alpha_{j,t})^2 \\ & + 2\eta\sigma_w^2\|\boldsymbol{L}\|_* + 4\sigma_w^2 \sum_t \sum_k |\mathcal{G}_{k,t}| \log(\theta_{k,t} + \epsilon) \\ & + \sigma_w^2\|\boldsymbol{A}\|_F^2, \end{aligned} \tag{13}$$

where the first term is the data term represented in the pixel-wise form, $\mathcal{G}_{k,t}$ denotes the $k$-th groups of the pixels of the $t$-th frame, $|\mathcal{G}_{k,t}|$ denotes the number of pixels in the group of $\mathcal{G}_{k,t}$, and $\theta_{k,t}$ denotes the standard deviation of the pixels belong to $\mathcal{G}_{k,t}$. By sharing the same Gaussian parameter $\theta_{k,t}$ over the group of pixels, the depedencies among the neighboring pixels can be exploited, leading to further improvement of the foreground estimation.

## IV. Optimization Algorithm

Similar to the standard RPCA problem, the proposed objective function of Eq. (13) can be solved by alternative updating the estimates of the background component $\boldsymbol{L}$ and the foreground component $\boldsymbol{S}$. For an initial estimate of $\boldsymbol{L}$, we solve for the foreground $\boldsymbol{S}$ by alternatively optimizing the positive multipliers $\boldsymbol{\Theta}$ and the Gaussian variables $\boldsymbol{A}$, and update $\boldsymbol{L}$ with fixed $\boldsymbol{S} = \boldsymbol{\Theta} \odot \boldsymbol{A}$.

### A. Solving the $\boldsymbol{S}$-Subproblem

Different from the standard RPCA problem, the estimate of the sparse component $\boldsymbol{S}$ is obtained by alternatively estimating the positive multipliers $\boldsymbol{\Theta}$ and the Gaussian variables $\boldsymbol{A}$. For a given $\boldsymbol{L}$, $\boldsymbol{S} = \boldsymbol{\Theta} \odot \boldsymbol{A}$ can be solved by minimizing

$$(\boldsymbol{\Theta}, \boldsymbol{A}) = \underset{\boldsymbol{\Theta},\boldsymbol{A}}{\operatorname{argmin}} \sum_t \sum_k \sum_{j \in \mathcal{G}_{k,t}} (D_{j,t} - L_{j,t} - \theta_{k,t}\alpha_{j,t})^2 + 4\sigma_w^2 \sum_t \sum_k |\mathcal{G}_{k,t}| \log(\theta_{k,t} + \epsilon) + \sigma_w^2 \|\boldsymbol{A}\|_F^2. \quad (14)$$

Note that solving Eq. (14) equals to solving a sequences of the following minimization problem for each frame, $t = 1, 2, \cdots, N$

$$(\boldsymbol{\theta}_t, \boldsymbol{\alpha}_t) = \underset{\boldsymbol{\theta}_t,\boldsymbol{\alpha}_t}{\operatorname{argmin}} \sum_k \sum_{j \in \mathcal{G}_{k,t}} (D_{j,t} - L_{j,t} - \theta_{k,t}\alpha_{j,t})^2 + 4\sigma_w^2 \sum_k |\mathcal{G}_{k,t}| \log(\theta_{k,t} + \epsilon) + \sigma_w^2 \|\boldsymbol{\alpha}_t\|_2^2, \quad (15)$$

which can be solved by alternatively updating the estimates of $\boldsymbol{\theta}_t$ and $\boldsymbol{\alpha}_t$.

*1) Solving the $\boldsymbol{\theta}_t$-Subproblem:* For fixed $\boldsymbol{\alpha}_t$, $\boldsymbol{\theta}_t$ can be estimated by minimizing

$$\boldsymbol{\theta}_t = \underset{\boldsymbol{\theta}_t}{\operatorname{argmin}} \sum_k \sum_{j \in \mathcal{G}_{k,t}} (r_{j,t} - \theta_{k,t}\alpha_{j,t})^2 + 4\sigma_w^2 \sum_k |\mathcal{G}_{k,t}| \log(\theta_{k,t} + \epsilon), \quad s.\ t.\ \theta_{k,t} \geq 0, \quad (16)$$

where $r_{j,t} = D_{j,t} - L_{j,t}$. Moreover, each $\theta_{k,t}$ can be solved independently as

$$\theta_{k,t} = \underset{\theta_{k,t}}{\operatorname{argmin}} \sum_{j \in \mathcal{G}_{k,t}} (r_{j,t} - \theta_{k,t}\alpha_{j,t})^2 + 4\sigma_w^2 |\mathcal{G}_{k,t}| \log(\theta_{k,t} + \epsilon), \quad s.\ t.\ \theta_{k,t} \geq 0. \quad (17)$$

Though Eq. (17) is non-convex, $\theta_{k,t}$ can be solved in closed-form by taking $\frac{f(\theta_{k,t})}{d\theta_{k,t}} = 0$, where $f(\theta_{k,t})$ denotes the right hand side of Eq. (17). It is easy to verify that the solution to Eq. (17) is given by

$$\theta_{k,t} = \begin{cases} 0, & \text{if } (b + a\epsilon)^2/a^2 - 4(c + b\epsilon)/a < 0 \\ z_k, & \text{otherwise}, \end{cases} \quad (18)$$

where $a = \sum_{j \in \mathcal{G}_{k,t}} \alpha_{j,t}^2$, $b = -\sum_{j \in \mathcal{G}_{k,t}} r_{j,t}\alpha_{j,t}$, $c = 2\sigma_w^2 |\mathcal{G}_{k,t}|$, and $z_k = \operatorname{argmin}_{\theta_{k,t}} \{f(0), f(\theta^*)\}$, where $\theta^*$ is a stationary point of $f(\theta_{k,t})$, i.e.,

$$\theta^* = -\frac{b + a\epsilon}{2a} + \sqrt{\frac{(b + a\epsilon)^2}{a^2} - \frac{4(c + b\epsilon)}{a}}. \quad (19)$$

*2) Solving the $\boldsymbol{\alpha}_t$-Subproblem:* For fixed $\theta_{k,t}$, $\boldsymbol{\alpha}_t$ can be solved by minimizing

$$\boldsymbol{\alpha}_t = \underset{\boldsymbol{\alpha}_t}{\operatorname{argmin}} \sum_k \sum_{j \in \mathcal{G}_{k,t}} (r_{j,t} - \theta_{k,t}\alpha_{j,t})^2 + \sigma_w^2 \|\boldsymbol{\alpha}_t\|_2^2, \quad (20)$$

which can be rewritten as

$$\boldsymbol{\alpha}_t = \underset{\boldsymbol{\alpha}_t}{\operatorname{argmin}} \|\boldsymbol{r}_t - \boldsymbol{\Gamma}\boldsymbol{\alpha}_t\|_2^2 + \sigma_w^2 \|\boldsymbol{\alpha}_t\|_2^2, \quad (21)$$

where $\boldsymbol{\Gamma} = diag(\theta_j) \in \mathbb{R}^{p \times p}$, wherein $\theta_j = \theta_{k,t}$ for $j \in \mathcal{G}_{k,t}$. Eq. (21) can be easily solved in a closed-form, i.e.,

$$\boldsymbol{\alpha}_t = (\boldsymbol{\Gamma}^T\boldsymbol{\Gamma} + \sigma_w^2 \boldsymbol{I})^{-1}\boldsymbol{\Gamma}^T\boldsymbol{r}_t, \quad (22)$$

where $\boldsymbol{I}$ denotes the identity matrix. Since $(\boldsymbol{\Gamma}^T\boldsymbol{\Gamma} + \sigma_w^2 \boldsymbol{I})$ is a diagonal matrix, Eq. (22) can be easily computed. After estimating $\boldsymbol{\theta}_t$ and $\boldsymbol{\alpha}_t$, $\boldsymbol{s}_t$ can be computed as $\boldsymbol{s}_t = \boldsymbol{\theta}_t \odot \boldsymbol{\alpha}_t$.

### B. Solving the $\boldsymbol{L}$-Subproblem

For an initial estimate of $\boldsymbol{S}$, the background component $\boldsymbol{L}$ can be obtained by solving

$$\boldsymbol{L} = \underset{\boldsymbol{L}}{\operatorname{argmin}} \|\boldsymbol{D} - \boldsymbol{S} - \boldsymbol{L}\|_F^2 + 2\eta\sigma_w^2 \|\boldsymbol{L}\|_*, \quad (23)$$

which is a low-rank matrix approximation problem and can be solved in a closed-form, i.e., the singular value thresholding (SVT) [48]. However, in each iteration of SVT, the singular value decomposition (SVD) has to be performed on the entire sample matrix, which is very slow and memory consuming. Thus, the updating of $\boldsymbol{L}$ using SVT is not suitable for practical application of background subtraction. In this paper, we adopt the stochastic optimization technique of [49] to solve the subproblem Eq.(23), which can significantly reduce the computational complexity and the memory consumption. The basic idea of the stochastic optimization algorithm is to express the nuclear norm of $\boldsymbol{U}$ whose rank is upper bounded by $r$ as

$$\|\boldsymbol{L}\|_* = \inf_{\boldsymbol{U} \in \mathbb{R}^{p \times r}, \boldsymbol{V} \in \mathbb{R}^{N \times r}} \{\frac{1}{2}(\|\boldsymbol{U}\|_F^2 + \|\boldsymbol{V}\|_F^2) \quad s.\ t.\ \boldsymbol{L} = \boldsymbol{U}\boldsymbol{V}^T\}. \quad (24)$$

Then the low-rank approximation problem is reformulated as a low-rank matrix factorization problem by treating $\boldsymbol{U} \in \mathbb{R}^{p \times r}$ as the basis of the low-dimensional subspace and $\boldsymbol{V} \in \mathbb{R}^{N \times r}$ as the transformation coefficients of $\boldsymbol{L}$ with respect to $\boldsymbol{U}$. With Eq.(24), the background component $\boldsymbol{L}$ can then be estimated by minimizing

$$(\boldsymbol{U}, \boldsymbol{V}) = \underset{\boldsymbol{U},\boldsymbol{V}}{\operatorname{argmin}} \|\boldsymbol{D} - \boldsymbol{U}\boldsymbol{V}^T - \boldsymbol{S}\|_F^2 + \eta\sigma_w^2 (\|\boldsymbol{U}\|_F^2 + \|\boldsymbol{V}\|_F^2), \quad (25)$$

which can be further reexpressed as

$$(\boldsymbol{v}_t, \boldsymbol{U}) = \underset{\boldsymbol{v}_t,\boldsymbol{U}}{\operatorname{argmin}} \sum_{t=1}^{N} \{\|\boldsymbol{d}_t - \boldsymbol{U}\boldsymbol{v}_t - \boldsymbol{s}_t\|_2^2 + \eta\sigma_w^2 \|\boldsymbol{v}_t\|_2^2\} + \eta\sigma_w^2 \|\boldsymbol{U}\|_F^2, \quad (26)$$

where $\boldsymbol{v}_t$ is the $t$-th column of $\boldsymbol{V}$, i.e., the transformation coefficients of the $t$-th background frame with respect to $\boldsymbol{U}$. Thus, we can be seen that each $\boldsymbol{v}_t$ can be solved individually based on an observed frame $\boldsymbol{d}_t$.

*1) Solving the $\boldsymbol{v}_t$-Subproblem:* For fixed $\boldsymbol{s}_t$ and $\boldsymbol{U}$, $\boldsymbol{v}_t$ can be estimated by minimizing

$$\boldsymbol{v}_t = \underset{\boldsymbol{v}_t}{\operatorname{argmin}} \|\boldsymbol{d}_t - \boldsymbol{U}\boldsymbol{v}_t - \boldsymbol{s}_t\|_2^2 + \eta\sigma_w^2 \|\boldsymbol{v}_t\|_2^2, \quad (27)$$

which admits a closed-form solution

$$\boldsymbol{v}_t = (\boldsymbol{U}^T\boldsymbol{U} + \eta\sigma_w^2 \boldsymbol{I})^{-1}\boldsymbol{U}^T(\boldsymbol{d}_t - \boldsymbol{s}_t). \quad (28)$$

**Algorithm 1** Robust Foreground Estimation via Structured GSM Modeling

- **Initialization**:
  (1) Set parameter $\eta$, $\sigma_w^2$, $\boldsymbol{A}_0 \in \mathbb{R}^{r\times r} = \mathbf{0}$ and $\boldsymbol{B}_0 \in \mathbb{R}^{p\times r} = \mathbf{0}$;
  (2) Initialize $\boldsymbol{U}^{(0)} \in \mathbb{R}^{p\times r}$ via Eq.(32).
- **Outer loop**: for $t = 1, 2, ......, N$, **do** {Access each frame}
  (1)**Inner loop** (solving Eqs. (15) and (27) ) for $j = 1, 2, \cdots, J$, **do**
    (a) Update $\boldsymbol{\theta}_t$ via Eq.(18);
    (b) Update $\boldsymbol{\alpha}_t$ via Eq.(22);
    (c) Update $\boldsymbol{s}_t = \boldsymbol{\theta}_t \odot \boldsymbol{\alpha}_t$;
    (d) Update $\boldsymbol{v}_t$ via Eq.(28).
  **End for**
  (2) estimating the background component $\boldsymbol{l}_t$.
    (a) Update $\boldsymbol{U}_t$ via Eq. (31);
    (b) Update $\boldsymbol{l}_t = \boldsymbol{U}_t \boldsymbol{v}_t$.
- **End for**

*2) Basis $\boldsymbol{U}$ Updating:* Instead of updating the basis $\boldsymbol{U}$ after solving for entire matrix $\boldsymbol{V}$, here we adopt the online basis updating strategy to update $\boldsymbol{U}$. In the $t$-th iteration, after estimating $\boldsymbol{v}_t$, the basis $\boldsymbol{U}$ can be updated by minimizing

$$\begin{aligned}\boldsymbol{U}^{(t)} &\triangleq \underset{\boldsymbol{U}}{\operatorname{argmin}}\ \frac{1}{t}\sum_{i=1}^{t}\frac{1}{2}\|\boldsymbol{d}_i - \boldsymbol{U}\boldsymbol{v}_i - \boldsymbol{s}_i\|_F^2 + \frac{\eta\sigma_w^2}{2t}\|\boldsymbol{U}\|_F^2 \\ &\triangleq \underset{\boldsymbol{U}}{\operatorname{argmin}}\ \frac{1}{2}Tr[\boldsymbol{U}^T(\boldsymbol{A}_t + \eta\sigma_w^2\boldsymbol{I})\boldsymbol{U}] - Tr(\boldsymbol{U}^T\boldsymbol{B}_t), \quad (29)\end{aligned}$$

where $\boldsymbol{A}_t$ and $\boldsymbol{B}_t$ in Eq.(29) are updated by

$$\begin{aligned}\boldsymbol{A}_t &\leftarrow \boldsymbol{A}_{t-1} + \boldsymbol{v}_t\boldsymbol{v}_t^T \\ \boldsymbol{B}_t &\leftarrow \boldsymbol{B}_{t-1} + (\boldsymbol{d}_t - \boldsymbol{s}_t)\boldsymbol{v}_t^T, \end{aligned} \quad (30)$$

with initialization $\boldsymbol{A}_0 = \mathbf{0}$ and $\boldsymbol{B}_0 = \mathbf{0}$. To solve Eq.(29), the block-coordinate descent with warm start $\boldsymbol{U}^{(t-1)}$ [50] is used, which allows each column of the basis $\boldsymbol{U}^{(t)}$ to be updated individually when the other columns are fixed. The $i$-th column $\boldsymbol{u}_i^{(t)}$ of $\boldsymbol{U}^{(t)}$ is updated as

$$\boldsymbol{u}_i^{(t)} \leftarrow \boldsymbol{u}_i^{(t-1)} + \frac{1}{\hat{\boldsymbol{A}}_{i,i}}(\boldsymbol{b}_i - \boldsymbol{U}^{(t-1)}\hat{\boldsymbol{a}}_i), \quad (31)$$

where $\hat{\boldsymbol{A}} = \boldsymbol{A}_t + \mu\boldsymbol{I}$, $\hat{\boldsymbol{a}}_i$ and $\boldsymbol{b}_i$ denote the $i$th column of $\hat{\boldsymbol{A}}$ and $\boldsymbol{B}_t$, respectively. The basis $\boldsymbol{U}_0$ is initialized by the bilateral random projections method [51]. Let $\boldsymbol{R}_1 \in \mathbb{R}^{n\times r}$ and $\boldsymbol{R}_2 \in \mathbb{R}^{m\times r}$ denote the two bilateral random projections constructed using the method of [51]. Then, $U_0$ can be initialized as

$$\boldsymbol{U}_0 = \boldsymbol{Y}_1(\boldsymbol{R}_2^T\boldsymbol{Y}_1)^{-1}\boldsymbol{Y}_2^T, \quad (32)$$

where $\boldsymbol{Y}_1 = \boldsymbol{Y}\boldsymbol{R}_1$, and $\boldsymbol{Y}_2 = \boldsymbol{Y}^T\boldsymbol{R}_2$, and $\boldsymbol{Y} \in \mathbb{R}^{m\times n}$ denotes matrix representation of the first frame.

The proposed foreground estimation via structured GSM modeling is then summarized in **Algorithm 1**. In the proposed algorithm, we estimate the foreground and background components individually for each frame. The updates of $\boldsymbol{\theta}_t$, $\boldsymbol{\alpha}$, $\boldsymbol{v}_t$ and $\boldsymbol{u}_i$ can all be performed in closed-form solutions. We found that **Algorithm 1** empirically converges even when the inner loop only executes one iteration (i.e., $J = 1$). Hence, the proposed **Algorithm 1** is very efficient.

## V. Experimental Results

In this section, we verify the performance of the proposed Structured GSM modeling based background subtraction method (denoted as SGSM-BS) on several real video sequences from the perception test images sequences (PITS)[1] [52] and the change detection dataset[2] [53], [54]. To show the effects of the pixel grouping on the performance of the SGSM-BS method, we have also implemented a baseline of the proposed SGSM-BS method, i.e., enforcing the pixels $s_{k,t}$ within a block to share the same regularization parameter $\theta_{k,t}$. To this end, we divide each frames into many non-overlapping blocks of size $m \times m$ ($m = 5$ in our implementation) and model the foreground pixels in a block with the same Gaussian parameter $\theta_{k,t}$. We denote this method as SGSM-BS-block. We have also implement the proposed GSM based method without considering the correlations among neighboring pixels, which solves the objective function of Eq.(11). Similar to Eq. (13), Eq.(11) can also be solved by alternative optimization. The major parameters of the proposed method are empirically set as: $r = 15$, $\eta = \frac{400}{\sqrt{p}}$, $\sigma_w^2 = 1.05 \times 10^{-3}$. Note that the parameters are fixed for all the test sequences, and the pixels of the test videos are normalized to the range of $[0, 1]$. To evaluate the performance of the proposed method, the criteria of $precision$ and $recall$ are employed:

$$precision = \frac{TP}{TP + FP}, \quad recall = \frac{TP}{TP + FN}, \quad (33)$$

where $TP$ (true positives) is the total numbers of pixels correctly classified as foreground, $FP$ (false positives) is the total numbers of pixels incorrectly classified as foreground, and $FN$ (false negatives) is the total numbers of foreground pixels incorrectly classified as background. As the harmonic mean of the $precison$ and $recall$, the $F\text{-}measure$ is employed:

$$F - measure = 2\frac{precision \cdot recall}{precision + recall}. \quad (34)$$

### A. Perception Test Images Sequences

The widely used Perception Test Images Sequences dataset contains 9 real sequences, which covers a variety of scenarios including static background with short-time lingering objects (Hall, Shoppimg Mall and Bootstrap), dynamic background (Water Surface, Fountain, Campus, Escalator and Curtain) and sudden illumination changes (Lobby), as shown in Fig.2. The frame numbers of the test videos are between 523 and 3584, and only 20 ground truth frames of the foreground parts are provided for evaluation.

To demonstrate the effectiveness of the proposed method, we compare the proposed method with several well-known methods, including the Detecting Contiguous outliers in

[1] perception.i2r.a-star.edu.sg/bk_model/bk_index.html
[2] www.changedetection.net

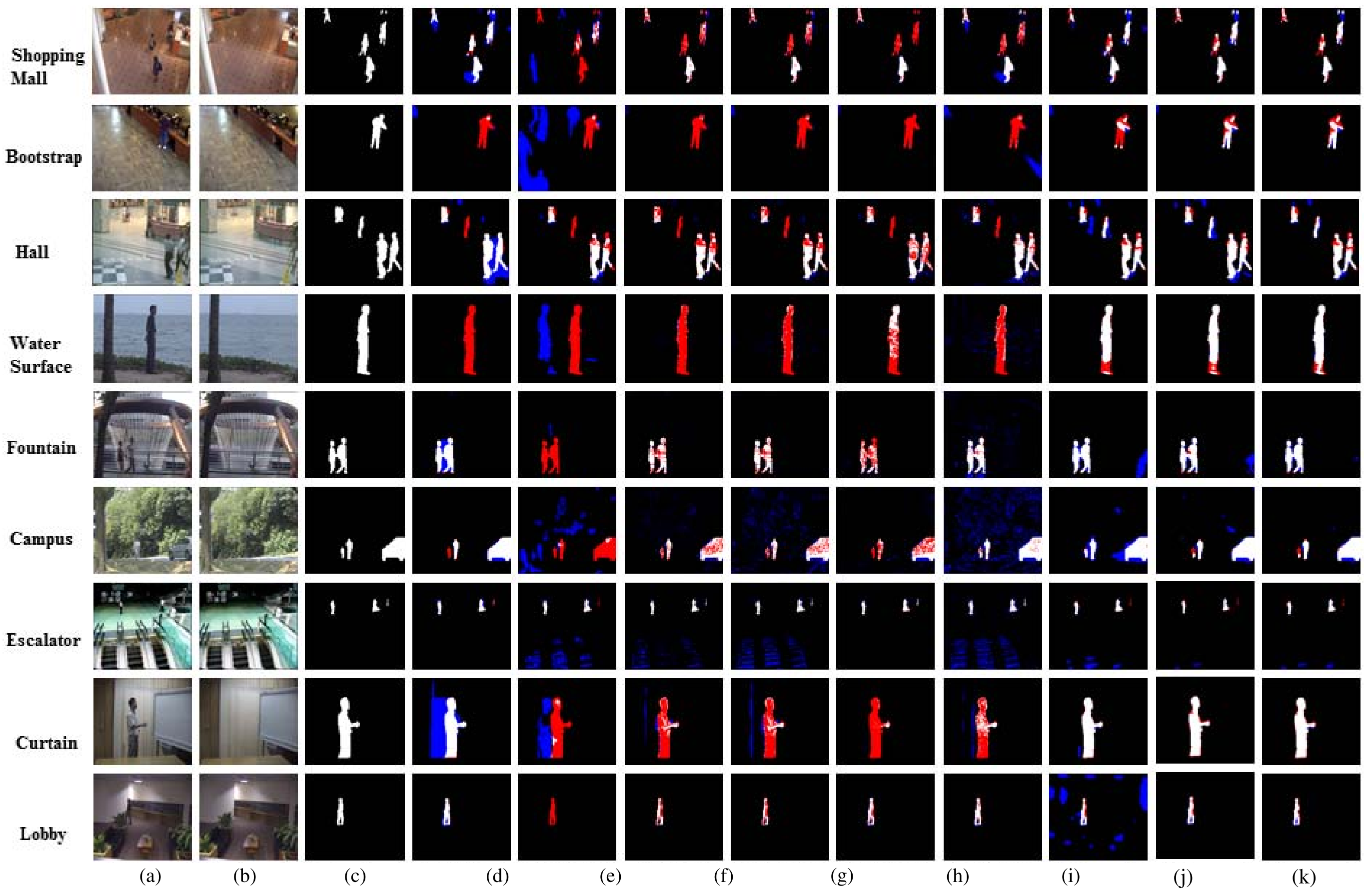



Fig. 2. Background subtraction results of 9 videos from PTIS dataset. (a) Original frame; (b) Background obtained by the proposed SGSM-BS method; (c) Groundtruth foreground mask; Foreground mask obtained from (d) DECOLOR [21]; (e) SAC [55]; (f) PCP [20]; (g) PRMF [57]; (h) PBAS [8]; (i) GMM [9]; (j) **Proposed GSM-BS**; (k) **Proposed SGSM-BS-block**; (l) **Proposed SGSM-BS**. (White represents correctly detected foreground, red represents missing pixels, and blue represents false alarm pixels).

the Low-Rank Representation (DECOLOR) method[3] [21], the Smoothness and Arbitrariness Constraints (SAC) method[4] [55], the Motion Saliency Detection (MSD) method[5] [56], the Principle Component Pursuit (PCP) method Component Pursuit (PCP) method the Probabilistic Robust Matrix Factorization (PRMF) method[6] [57], and several recently proposed state-of-the-art RPCA based methods: Low-Rank and[7] [57], and several recently proposed state-of-the-art RPCA based methods: Low-Rank and Structured Spare Decomposition (LRSSD) method [23], the TV regularized RPCA (TVRPCA) method [24], and the Modified Linear Regression with Basis Selection (MLRBS) method [58]. Two pixel-based methods, i.e., Pixel-Based Adaptive Segmenter (PBAS)[8] [8] and the improved Gaussina Mixture Model (GMM) [9] [9] are also included.

Fig.2 shows the visual comparison results of the detected foreground masks on the PTIS dataset. As shown in the second column of Fig.2, accurate backgrounds can be estimated by the proposed SGSM-BS method. For the "Shopping Mall" and "Bootstrap" scenarios, which show an indoor environment where people keep walking in most cases but a man lingers for a while, the SAC, PCP, PRMF, PBAS, and the GMM methods fail to accurately detect the lingering man. The DECOLOR and the proposed methods perform better than these methods for these two scenarios. For the "Hall" and "Water surface" scenarios, which are more challenging as there are long-time lingering objects, only the proposed GSM-BS and SGSM-BS methods can detect the long-time lingering woman or man, and other methods classified the long-time lingering objects as the background. Compared with the GSM-BS, the proposed SGSM-BS method can much reduce the false positive or negative points. Though all the methods can successfully detect the moving objects for the "Fountain" and "Campus" scenarios, the proposed GSM-BS and SGSM-BS methods obviously produce much less false positives. For the "Escalator" scenario, only the proposed SGSM-BS, DECOLOR and PBAS methods can handle the dynamic background caused by the motion of the escalator. For the "Curtain" scenario, which shows a man wearing a white color shirt whose color is similar to the fluttered curtain caused by the wind, only the DECOLOR, and the proposed GSM-BS and SGSM-BS methods can detect the white color

[3]https://fling.seas.upenn.edu/∼xiaowz/dynamic/wordpress/my-uploads/codes/decolor.zip
[4]http://cs.tju.edu.cn/orgs/vision/∼xguo/code/RFDSA_ECCV14.rar
[5]http://cs.tju.edu.cn/orgs/vision/msd/Motion_Detection.zip
[6]http://winsty.net/prmf/code.zip
[7]http://winsty.net/prmf/code.zip
[8]https://sites.google.com/site/pbassegmenter/download-1
[9]http://www.zoranz.net/Publications/CvBSLibGMM.zip

TABLE I

PERFORMANCE OF *F-Measure* (%) ON PERCEPTION TEST IMAGES SEQUENCES DATASET

| Method \ Video | Shopping Mall | Lobby | Curtain | Fountain | Campus | Hall | Bootstrap | Escalator | Water Surface | Average |
|---|---|---|---|---|---|---|---|---|---|---|
| DECOLOR [21] | 68.06 | 56.52 | 81.58 | 82.76 | 73.29 | 53.20 | 56.86 | <u>75.46</u> | 79.65 | 69.71 |
| SAC [55] | 74.07 | 80.29 | 89.76 | 75.44 | 67.79 | 66.73 | 68.41 | 63.53 | 87.96 | 74.89 |
| MSD [56] | 73.34 | 46.88 | 30.25 | 63.31 | 66.75 | 49.87 | 54.62 | 66.53 | 27.74 | 53.25 |
| PCP [20] | 66.78 | 62.18 | 81.28 | 63.74 | 40.69 | 50.25 | 56.78 | 63.54 | 39.76 | 58.33 |
| PRMF [57] | 69.68 | 52.13 | 58.94 | 53.80 | 36.58 | 58.88 | 59.62 | 37.79 | 35.40 | 51.42 |
| LRSSD [23] | 73.62 | 73.13 | 83.57 | 83.71 | 76.13 | 72.22 | 58.42 | 72.14 | 90.50 | 75.93 |
| MLRBS [58] | 74.74 | 70.93 | 87.25 | <u>86.52</u> | 70.17 | 67.64 | 69.72 | **76.22** | 87.24 | 76.72 |
| TVRPCA [24] | – | 75.00 | – | 80.00 | 77.00 | 69.00 | 69.00 | 66.00 | 88.00 | – |
| PBAS [8] | 71.77 | 56.74 | 72.48 | 62.04 | 69.50 | 61.92 | 55.38 | 31.13 | 89.02 | 63.33 |
| GMM [9] | 67.25 | 42.62 | 30.52 | 50.06 | 34.77 | 42.17 | 66.36 | 50.44 | 32.88 | 46.34 |
| **Proposed GSM-BS** | <u>79.28</u> | 52.97 | <u>92.35</u> | 67.54 | 64.68 | <u>76.28</u> | 78.43 | 66.15 | <u>91.78</u> | 74.37 |
| **Proposed SGSM-BS-block** | 77.70 | <u>83.78</u> | 79.12 | 86.44 | <u>77.93</u> | 74.96 | **79.05** | 69.11 | 71.21 | <u>77.70</u> |
| **Proposed SGSM-BS** | **79.89** | **86.02** | **93.54** | **87.16** | **83.15** | **77.59** | <u>78.75</u> | 71.70 | **92.88** | **83.41** |

TABLE II

PERFORMANCE OF *F-Measure* (%) ON DYNAMIC BACKGROUNDS CATEGORY OF THE 2014 CD DATASET

| Method \ Video | boats | canoe | fall | fountain01 | fountain02 | overpass | Average |
|---|---|---|---|---|---|---|---|
| DECOLOR [21] | 49.48 | 6.67 | 54.20 | 2.71 | 75.22 | <u>87.72</u> | 46.00 |
| PCP [20] | 34.15 | 37.09 | 34.90 | 4.31 | 45.92 | 47.92 | 34.05 |
| LRSSD [23] | 79.71 | 83.51 | 74.27 | 33.15 | 85.36 | 74.39 | 71.73 |
| MLRBS [58] | 81.61 | 82.89 | 73.60 | 36.66 | <u>94.17</u> | 88.49 | 76.24 |
| TVRPCA [24] | 39.00 | 86.00 | 51.00 | 12.00 | 72.00 | 77.00 | 61.00 |
| SuBSENSE [59] | 69.32 | 79.23 | <u>86.61</u> | **75.31** | **94.41** | 85.72 | 81.70 |
| SOBS [10] | **89.57** | **95.24** | 27.75 | 11.58 | 81.77 | 88.37 | 65.71 |
| PBAS [8] | 36.11 | 71.96 | **87.14** | 41.73 | 93.55 | 79.25 | 68.29 |
| GMM [9] | 74.74 | 81.51 | 42.38 | 8.15 | 79.17 | 42.17 | 54.68 |
| **Proposed GSM-BS** | 88.05 | 91.49 | 85.04 | 63.74 | 79.35 | **88.56** | 82.71 |
| **Proposed SGSM-BS-block** | <u>88.81</u> | 88.03 | 82.71 | 66.70 | 84.61 | 86.30 | <u>82.86</u> |
| **Proposed SGSM-BS** | 88.73 | <u>92.39</u> | 86.44 | <u>67.66</u> | 87.77 | 85.96 | **84.83** |

shirt. But the DECOLOR method produces more false positives than the proposed SGSM-BS method due to the over-smooth MRF prior. For the "Lobby" scenario, which shows a scene with lighting changing, the SAC, PCP and PRMF methods fail to accurately estimate the compact mask of a man in the dark room. The DECOLOR, PBAS, GMM and the proposed SGSM-BS methods can obtain satisfactory results. Moreover, the proposed SGSM-BS method produces less false positives than GSM-BS, which further verifies the effectiveness of the structured GSM modeling. Overall, compared with other competing methods, the proposed SGSM-BS method achieves high true positives with small false positives.

Table I shows the *F-measure* results of the test methods on the PTIS dataset, from which one can see that in average the proposed method outperforms other competitive methods. The average *F-measure* gain over the state-of-the-art MLRBS method, which ranks the second in this comparison group, can be up to 6.7%.

### *B. Change Detection Dataset*

As one of the most difficult detection benchmark dataset, the 2014 Change Detection (2014 CD) dataset has been proposed in [54]. In the 2014 CD dataset, more than 70,000 frames have been captured and manually annotated. We have conducted experiments on all the categories of the CD dataset, except the PTZ category. As the videos of PTZ category were captured by the pan-tilt-zoom cameras and zooming mode, the backgrounds of the videos of PTZ category do not have the low-rank property, and thus the low-rank and sparse based methods (including our method) fail to deal with those videos. Here, we first present the comparison studies on three typical categories and then report the average results on all the categories except the PTZ category.

*1) Dynamic Background:* The dynamic background category contains 6 videos of the outdoor scenes with strong background motion. The frame numbers of the videos are from 1,189 to 8,000 frames. Table II reports the *F-measure* results of the test methods, where the results of the other competing methods are obtained from their papers or the 2014 CD dataset. For DECOLOR and PCP methods, we reproduce the results using their source codes with default parameters released by the authors. From Tabel II, we can see that the proposed SGSM-BS method achieves the best foreground detection performances. Fig. 3 shows some visual comparison results of the test methods on this category. It can be seen that the batch-based methods, e.g., the DECOLOR and PCP methods fail to recover the complete mask of the moving objects for the "boats" and "canoe" scenarios, where the objects moving slowing in a period of time can be considered as backgrounds. For the "fall", "fountain01"and "fountain02" scenarios, it is rather challenging to detect the foregrounds due to the significantly dynamic background, e.g., the leaves shaken by the wind and the fountain. Clearly, the proposed SGSM-BS methods (as well as its baseline SGSM-BS-block) achieve the best results for these scenarios. The proposed methods can perfectly detect the moving objects, demonstrating the effectiveness of the proposed GSM modeling techniques. It can be seen that the proposed SGSM-BS-block

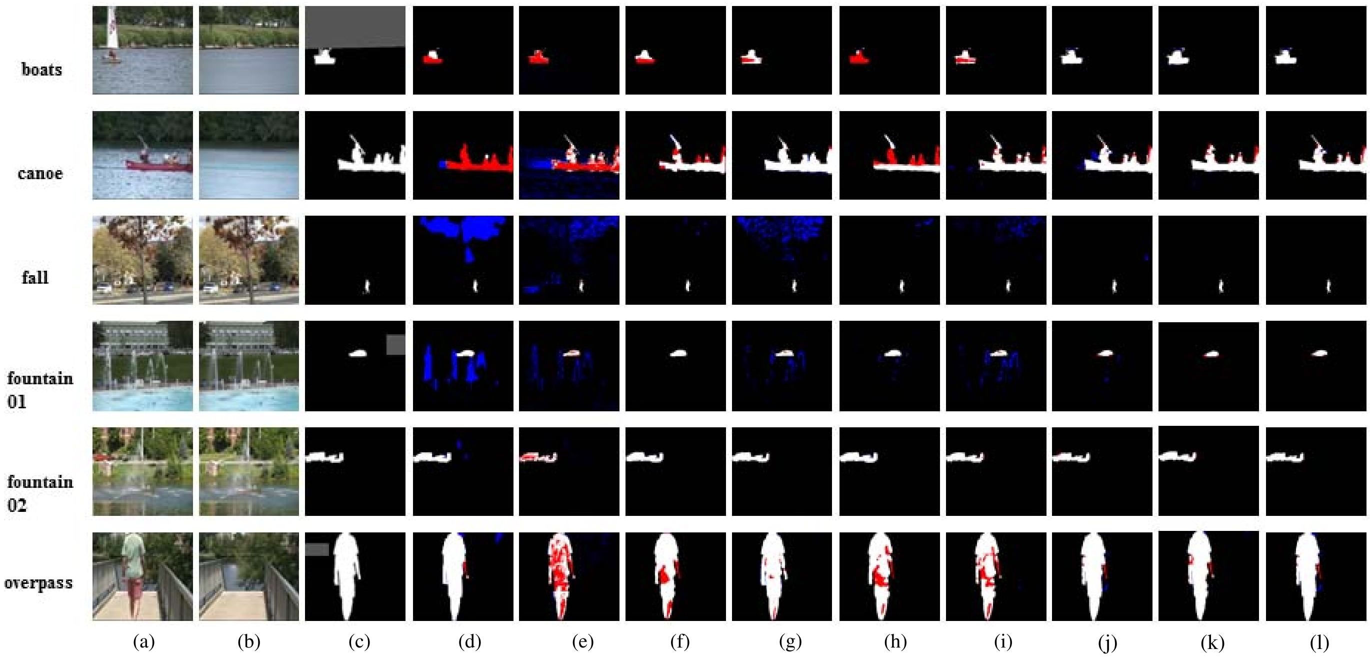



Fig. 3. Background subtraction results of 6 videos from Dynamic Backgroud category of 2014 CD dataset. (a) Original frame; (b) Background obtained from the proposed SGSM-BS method; (c) Groundtruth foreground mask; Foreground mask obtained from (d) DECOLOR [21]; (e) PCP [20]; (f) SuBSENSE [59]; (g) SOBS[10]; (h) PBAS [8]; (i) GMM [9]; (j) **Proposed GSM-BS**; (k) **Proposed SGSM-BS-block**; (l) **Proposed SGSM-BS**. (White represents correctly detected foreground, red represents missing pixels, and blue represents false alarm pixels).

TABLE III

PERFORMANCE OF *F-Measure* (%) ON INTERMITTENT OBJECT MOTION CATEGORY OF THE 2014 CD DATASET

| Method \ Video | abandonedBox | parking | sofa | streetLight | tramstop | winterDriveway | Average |
|---|---|---|---|---|---|---|---|
| DECOLOR [21] | 46.80 | 22.67 | 52.29 | 30.45 | 53.80 | 71.66 | 46.28 |
| PCP [20] | 30.31 | 30.79 | 42.26 | 16.66 | 36.34 | 32.91 | 31.55 |
| SuBSENSE [59] | 84.95 | 44.38 | 74.24 | **98.83** | 55.55 | 36.20 | 65.70 |
| SOBS [10] | 56.73 | 40.34 | 64.03 | 97.29 | 84.18 | 12.15 | 59.12 |
| PBAS [8] | 36.11 | 71.96 | 87.14 | 41.73 | **93.55** | **79.25** | 68.29 |
| GMM [9] | 52.63 | 71.79 | 65.24 | 49.65 | 57.81 | 22.39 | 53.25 |
| **Proposed GSM-BS** | 95.61 | 45.18 | 87.20 | 92.54 | 65.58 | 26.89 | 68.83 |
| **Proposed SGSM-BS-block** | 87.86 | 65.65 | 67.79 | 84.93 | 72.38 | 43.33 | 70.32 |
| **Proposed SGSM-BS** | **96.35** | **79.66** | **87.28** | 97.33 | 91.94 | 41.62 | **81.86** |

performs mostly better than the GSM-BS method that does not exploit the spatial correlations among the neighboring pixels. However, due to the inaccurate grouping of the pixels, the performance of the SGSM-BS-block method is inferior to that of the SGSM-BS method.

*2) Intermittent Object Motion:* The intermittent object motion category contains 6 videos with scenarios known for causing the ghosting artifacts. Durations of the videos are from 2,500 to 4,500 frames. This category is adopted for testing the adaptivity of the test methods to the changing background. We compared the proposed method only with the DECOLOR, PCP, SuBSENSE, SOBS, PBAS, and GMM methods, as the results of LRSSD, MLRBS and TVRPCA methods are unavailable in their papers and their codes are also unavailable online. As shown in Table III, on average the proposed GSM-BS and SGSM-BS methods achieved the best performance for this category. For the "abandon Box", "sofa" and "tramstop" scenarios, batch-based methods such as DECOLOR and PCP methods fail to detect the boxes, which are abandoned for a period of time. Though the SuBSENSE, SOBS, PBAS and GMM methods can detect these boxes, they missed many details of the foregrounds and produced false detections. In contrast, the proposed SGSM-BS, SGSM-BS-block and GSM-BS methods can successfully detect these boxes. For the "winter Driveway", "parking" and "street Light" scenarios, the proposed methods also obtained the best estimations of the moving objects.

*3) Bad Weather:* The bad weather category contains 4 outdoor videos depicting the low-visibility winter storm conditions. Durations of the videos are from 3,500 to 7,000 frames. In this category, we compared the proposed methods with the DECOLOR, PCP, SuBSENSE, SOBS, and GMM methods, for which we can regenerate their results with the source codes or borrow the results from their papers. Table IV shows the *F-measure* results of the test methods, from which we can see that the proposed SGSM-BS method achieved the impressive performance. As shown in Fig. 5, the PCP, SOBS and GMM methods fail to detect the moving objects due to the low-visibility caused by the storm. Compared to these methods, the DECOLOR method performed much better by

TABLE IV
PERFORMANCE OF $F$-$Measure$ (%) ON BAD WEATHER CATEGORY OF THE 2014 CD DATASET

| Method \ Video | billzard | skating | snowFall | wetSnow | Average |
|---|---|---|---|---|---|
| DECOLOR [21] | 85.19 | 93.36 | 81.87 | 66.78 | 81.80 |
| PCP [20] | 81.92 | 84.47 | 77.04 | 59.66 | 75.77 |
| SuBSENSE [59] | 85.10 | 90.54 | **89.37** | **79.73** | **86.18** |
| SOBS [10] | 59.37 | 88.85 | 65.19 | 50.79 | 66.05 |
| GMM [9] | 74.74 | 81.51 | 42.38 | 57.62 | 54.68 |
| **Proposed GSM-BS** | 78.46 | 93.02 | 84.79 | 62.59 | 79.72 |
| **Proposed SGSM-BS-block** | 82.12 | 91.10 | 83.14 | 62.49 | 79.71 |
| **Proposed SGSM-BS** | **85.45** | **93.40** | 86.14 | 77.24 | 85.56 |

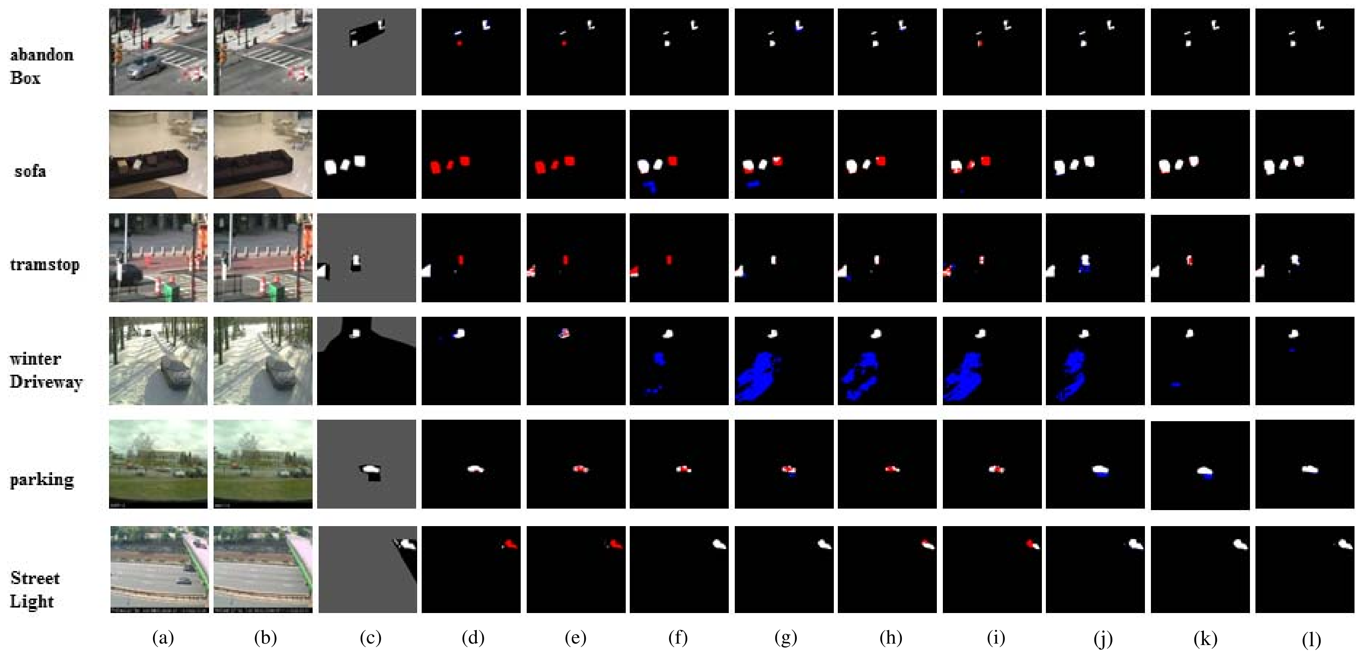



Fig. 4. Background subtraction results of 6 videos from Intermittent Object Motion category of 2014 CD dataset. (a) Original frame; (b) Background obtained from the proposed SGSM-BS method; (c) Groundtruth foreground mask; Foreground mask obtained from (d) DECOLOR [21]; (e) PCP [20]; (f) SuBSENSE [59]; (g) SOBS [10]; (h) PBAS [8]; (i) GMM [9]; (j) **Proposed GSM-BS**; (k) **Proposed SGSM-BS-block**; (l) **Proposed SGSM-BS**. (White represents correctly detected foreground, red represents missing pixels, and blue represents false alarm pixels).

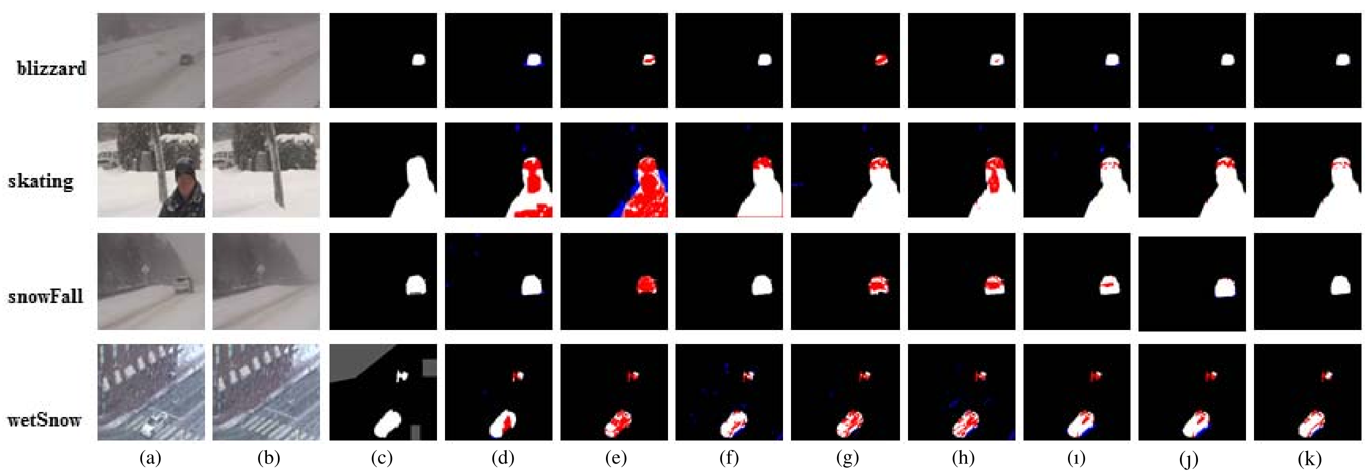


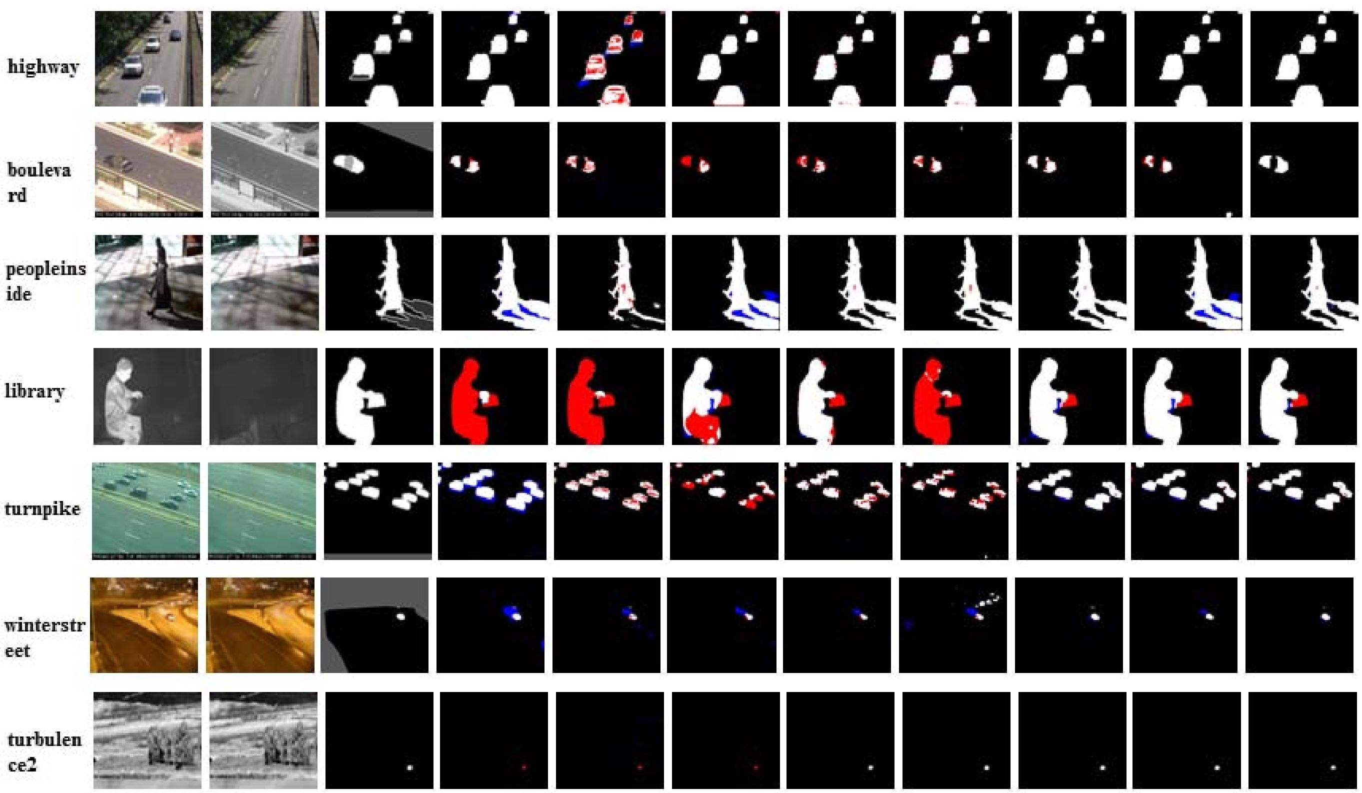

Fig. 5. Background subtraction results of 4 videos from Bad Weather category of 2014 CD dataset. (a) Original frame; (b) Background obtained from the proposed SGSM-BS method; (c) Groundtruth foreground mask; Foreground mask obtained from (d) DECOLOR [21]; (e) PCP [20]; (f) SuBSENSE [59]; (g) SOBS [10]; (h) GMM [9]; (i) **Proposed GSM-BS**;(j) **Proposed SGSM-BS-block**; (k) **Proposed SGSM-BS**. (White represents correctly detected foreground, red represents missing pixels, and blue represents false alarm pixels).

exploiting the low-rank and sparse properties. Without exploiting the pixel correlations, the proposed GSM-BS method produced some false alarm and the accuracy was slightly inferior to the DECOLOR method. By exploiting the neighboring pixels correlations, the proposed SGSM-BS method further improved the performance and produces the best results.

TABLE V

AVERAGE PERFORMANCE OF *F-Measure* (%) ON 10 CATEGORIES OF THE 2014 CD DATASET

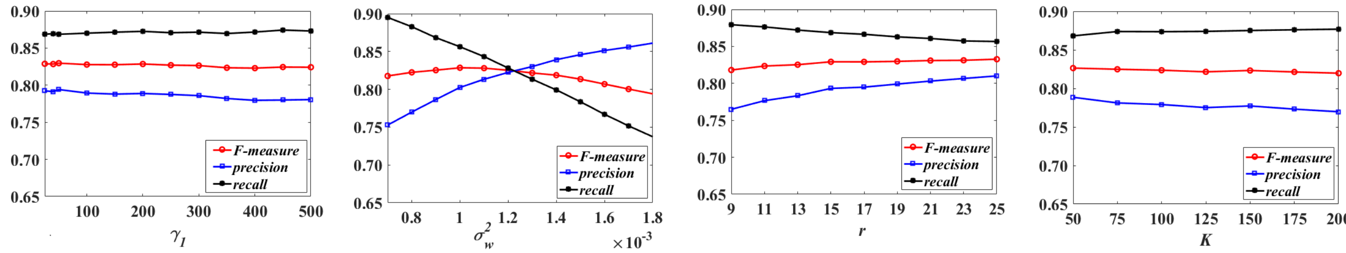
| Category / Method | Baseline | Dynamic Background | Camera Jitter | Shadows | Thermal | Intermittent Object Motion | Bad weather | Low Framerate | Night Videos | turbulence | $Avg_7$ | $Avg_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LRSSD [23] | 92 | 71 | 78 | 81 | 75 | 67 | 70 | – | – | – | 77 | – |
| TVRPCA [24] | 84 | 55 | 63 | 71 | 69 | 57 | 78 | – | – | – | 68 | – |
| 2P-RPCA [22] | 92 | 78 | 81 | 80 | 76 | 65 | 75 | – | – | – | 78 | – |
| LR-FSO [60] | 80 | 74 | 76 | 60 | 80 | 63 | 79 | – | – | – | 74 | – |
| GFL [27] | 83 | 74 | 78 | 82 | 76 | 59 | 76 | – | – | – | 75 | – |
| GRASTA [61] | 66 | 35 | 43 | 52 | 42 | 35 | 68 | – | – | – | 48 | – |
| BMTDL [62] | 88 | 75 | 72 | 81 | 79 | 69 | 77 | – | – | – | 77 | – |
| RMAMR [25] | 89 | 82 | 75 | 73 | 75 | 66 | 70 | – | – | – | 75 | – |
| SRPCA [31] | 82 | 84 | 78 | 77 | 79 | 80 | 75 | – | – | – | 79 | – |
| MSCL-FL [32] | 94 | **90** | **86** | 86 | **86** | **84** | **88** | – | – | – | **88** | – |
| PBAS [8] | 92 | 68 | 72 | 85 | 75 | 68 | – | – | – | – | – | – |
| DECOLOR [21] | 92 | 46 | 67 | 83 | 70 | 46 | 82 | 50 | 42 | 74 | 69 | 65 |
| PCP [20] | 60 | 34 | 54 | 51 | 34 | 32 | 76 | 44 | 43 | 50 | 49 | 48 |
| SuBSENSE [59] | **95** | 81 | 81 | **89** | 81 | 65 | 86 | 65 | 49 | 84 | 83 | 78 |
| SOBS [10] | 92 | 65 | 82 | 82 | 68 | 59 | 66 | 55 | 38 | 57 | 73 | 66 |
| GMM [9] | 91 | 54 | 56 | 73 | 65 | 53 | 54 | 50 | 37 | 55 | 64 | 59 |
| **Proposed GSM-BS** | 93 | 83 | 81 | 87 | 82 | 69 | 80 | 72 | 47 | 82 | 80 | 78 |
| **Proposed SGSM-BS-block** | 93 | 83 | 81 | 86 | 82 | 70 | 80 | 73 | 49 | 82 | 82 | 80 |
| **Proposed SGSM-BS** | **95** | 85 | 82 | **89** | 85 | 82 | 86 | **75** | **51** | **85** | 86 | **82** |




(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

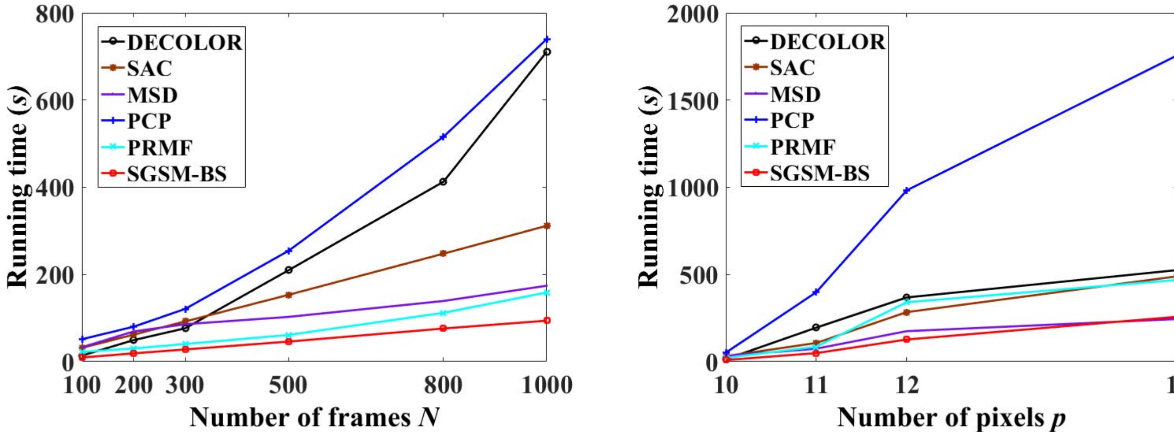
Fig. 6. Background subtraction results from 7 categories of 2014 CD dataset. (a) Original frame; (b) Background obtained from the proposed SGSM-BS method; (c) Groundtruth foreground mask; Foreground mask obtained from (d) DECOLOR [21]; (e) PCP [20]; (f) SuBSENSE [59]; (g) SOBS [10]; (h) GMM [9]; (i) **Proposed GSM-BS**; (j) **Proposed SGSM-BS-blcok**; (k) **Proposed SGSM-BS**. (White represents correctly detected foreground, red represents missing pixels, and blue represents false alarm pixels).

*4) Average Result:* We have also conducted experiments on other categories (except the PTZ) of the 2014 CD dataset. We have also compared the proposed method with 2P-RPCA [22], LR-FSO [60], GFL [27], GRASTA [61], BMTDL [62], RMAMR [25], SRPCA [31] and the current state-of-the-art method MSCL-FL [32]. For fair comparisons, the results of other competing methods were obtained directly from their papers or the 2014 CD dataset website, except the DECOLOR and PCP methods, for which we generated the results using their source codes (with the default parameters) released by the authors. Table V shows the average *F-measure* results of the test methods on the 2014 CD dataset. In Table V, $Avg_7$




(a) (b) (c) (d)

Fig. 7. The *F-measure*, *precision*, and *recall* curves as functions of (a) the regularization parameter $\gamma_1$, (b) the regularization parameter $\sigma_w^2$, (c) the number of bilateral random projections $r$, and the number of superpixels $K$.




(a) (b)

Fig. 8. The running time (seconds) curves as functions of the number of frames $N$, and the number of pixels $p$ in log domain. (a) The running time (seconds) as a function of the number of frames $N$; (b) The running time (seconds) as a function of the number of pixels $p$ in log domain.

denotes the averaged *F-measure* results on the 7 categories, while $\text{Avg}_{10}$ denotes the averaged *F-measure* results on all the 10 categories. From Table V, we can see that the proposed SGSM-BS method outperforms most of the other competing methods, and is comparable with the MSCL-FL method that is the current state-of-the-art method. Fig. 6 shows some visual comparison results of the competing methods on several videos of the 2014 CD dataset. It can be seen that the proposed SGSM-BS method performs comparably with or even better than other methods.

### C. Parameters Selection

In the proposed method, there are only four free parameters needed to be tuned, i.e., the regularization parameter $\eta = \frac{\gamma_1}{\sqrt{p}}$, the variance of the approximation error $\sigma_w^2$, the bilateral random projections number $r$ and the number of superpixels $K$. Fig.7 shows the average *F-measure*(%), *precision*(%) and *recall*(%) results curves as functions of $\gamma_1$, $\sigma_w^2$, $r$ and $K$ on the Perception Test Images Sequences dataset. From Fig.7(a) it can be seen that the performance of the proposed method is insensitive to the parameter $\gamma_1$. In our implementation, we set $\eta = \frac{400}{\sqrt{p}}$. From Fig.7(b), one can see that the precision result increases and the recall result decreases when increasing the value of $\sigma_w^2$. In general, the performance of the proposed method is robust to the values of $\sigma_w^2$ in the range of $[0.9, 1.2] \times 10^{-3}$. In our implementation, we set $\sigma_w^2 = 1.05 \times 10^{-3}$. From Fig.7(c) it can be seen that the performance of the proposed method is also insensitive to the bilateral random projections number $r$. For the tradeoff between the performance and the computational complexity, we set $r = 15$ in our implementation. We have also conducted experiments to verify the effects of the number of superpixels on the performance of the proposed SGSM-BS method. Fig.7(d) shows the curves of the average *F-measure*(%), *precision*(%) and *recall*(%) as functions of the parameter $K$ on the Perception Test Images Sequences dataset. It can be seen that the performance of the proposed SGSM-BS method is insensitive to the values of $K$. In our implementation, we set $K = 100$.

### D. Computational Complexity

The computational complexity of the proposed method in **Algorithm 1** mainly consists of three parts: 1) estimating $\boldsymbol{s}_t$; 2) estimating $\boldsymbol{v}_t$; 3) updating $\boldsymbol{U}_t$. The complexity of estimating $\boldsymbol{s}_t$ is $O(p)$, as the complexity of updating $\boldsymbol{\theta}_t$ and $\boldsymbol{\beta}_t$ are both $O(p)$. The complexity of estimating $\boldsymbol{v}_t$ is $O(pr^2)$. The complexity of updating $\boldsymbol{U}_t$ is $O(pr^2)$. Thus, the overall complexity of the proposed methods for a $N$ frames video is $O(Npr^2)$, which is linear in the resolution of the frame and number of frames. In contrast, the batch-based PCP method requires SVD operations, whose complexity is $O(Np^2)$. Since $p \gg r^2$, our method is much faster than the batch-based PCP methods. The proposed method was implemented with Matlab language on an Intel Core i7-3770 3.4GHz CPU. Fig.8(a) shows the running time curve as a function of the data size $N$ when the resolution of the frame $p$ is fixed as 20480. Fig.8(b) shows the running time curve as a function of the resolution of the frame $p$ in log domain when the data size $N$ is fixed as 100. From Fig.8(a), we can see the running time of the proposed method is linear with the data size. The PCP method is slowest as the SVD operation is very time-consuming. It can be seen that the proposed method runs much faster than the DECOLOR method and the MSD method.

In addition to the running time, the memory cost is also significantly reduced. The memory required by batch-based PCP method is $O(pN)$, whereas the memory required by the proposed method is $O(pr)$, which is much smaller than that of the PCP method.

## VI. Conclusion

Though the RPCA framework has been successfully used for background subtraction, the conventional $\ell_1$ norm based sparse regularizer cannot characterize the foreground components of varying sparsity. In this paper, we propose to model the sparse foregrounds with Gaussian Scale Mixture (GSM) models. Compared with the $\ell_1$ norm, the GSM model based sparse regularizer can jointly estimating the regularizer parameters and the unknown sparse coefficients from the observed video, leading to significant improvements. Moreover, we further extend the GSM models to structured GSM models by considering the correlations between the neighboring pixels. By modeling the pixels within each homogeneous regions with the same GSM model, the foreground estimation accuracy can be further improved. Experimental results on several challenging scenarios show that the proposed method performs much better than most of existing background subtraction methods in terms of both performance and speed.

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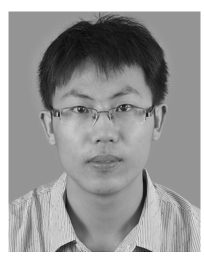
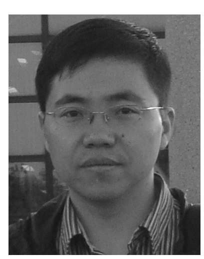
**Guangming Shi** (SM'10) received the B.S. degree in automatic control, the M.S. degree in computer control, and the Ph.D. degree in electronic information technology from Xidian University in 1985, 1988, and 2002, respectively.

He joined the School of Electronic Engineering, Xidian University, in 1988. From 1994 to 1996, he was with the Department of Electronic Engineering, The University of Hong Kong, as a Research Assistant. Since 2003, he has been a Professor with the School of Electronic Engineering, Xidian University, and in 2004, he was the Head of National Instruction Base of Electrician and Electronic. In 2004, he had studied with the Department of Electronic Engineering, University of Illinois at Urbana–Champaign. Since 2018, he has been a Professor with the School of Artificial Intelligence, Xidian University. He is currently the Deputy Director of the School of Electronic Engineering, Xidian University, and the Academic Leader in the subject of circuits and systems. He has authored or co-authored over 60 research papers. His research interests include compressed sensing, theory and design of multirate filter banks, image denoising, low-bit-rate image/video coding, and implementation of algorithms for intelligent signal processing (using DSP and FPGA).



**Tao Huang** is currently pursuing the Ph.D. degree with the School of Artificial Intelligence, Xidian University, Xi'an, China. His research interests include image processing, sparse representation, low rank matrix recovery, and deep learning.



**Weisheng Dong** (M'11) received the B.S. degree in electronic engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2004, and the Ph.D. degree in circuits and system from Xidian University, Xi'an, China, in 2010. He was a Visiting Student with Microsoft Research Asia, Bejing, China, in 2006. From 2009 to 2010, he was a Research Assistant with the Department of Computing, The Hong Kong Polytechnic University, Hong Kong. In 2010, he joined the School of Electronic Engineering, Xidian University, as a Lecturer, where he has been a Professor since 2016. Since 2018, he has been a Professor with the School of Artificial Intelligence, Xidian University. His research interests include inverse problems in image processing, sparse signal representation, and image compression.

He was a recipient of the Best Paper Award at the SPIE Visual Communication and Image Processing in 2010. He is currently serving as an Associate Editor of the IEEE TRANSACTIONS ON IMAGE PROCESSING.

**Jinjian Wu** received the B.Sc. and Ph.D. degrees from Xidian University, Xi'an, China, in 2008 and 2013, respectively. From 2011 to 2013, he was a Research Assistant with Nanyang Technological University, Singapore. From 2013 to 2014, he was a Post-Doctoral Research Fellow with Nanyang Technological University. From 2013 to 2015, he was a Lecturer with Xidian University. From 2015 to 2018, he has been an Associate Professor with the School of Electronic Engineering, Xidian University. Since 2018, he has been an Associate Professor with the School of Artificial Intelligence, Xidian University. His research interests include visual perceptual modeling, saliency estimation, quality evaluation, and just noticeable difference estimation. He has served as a TPC member for ICME2014-2015, PCM2015-2016, ICIP2015, and QoMEX2016. He awarded the Best Student Paper of ISCAS 2013.

**Xuemei Xie** received the M.S. degree in electronic engineering from Xidian University in 1994 and the Ph.D. degree in electrical and electronic engineering from The University of Hong Kong in 2004. Her research interests are digital signal processing, multirate filter banks, and wavelet transforms.